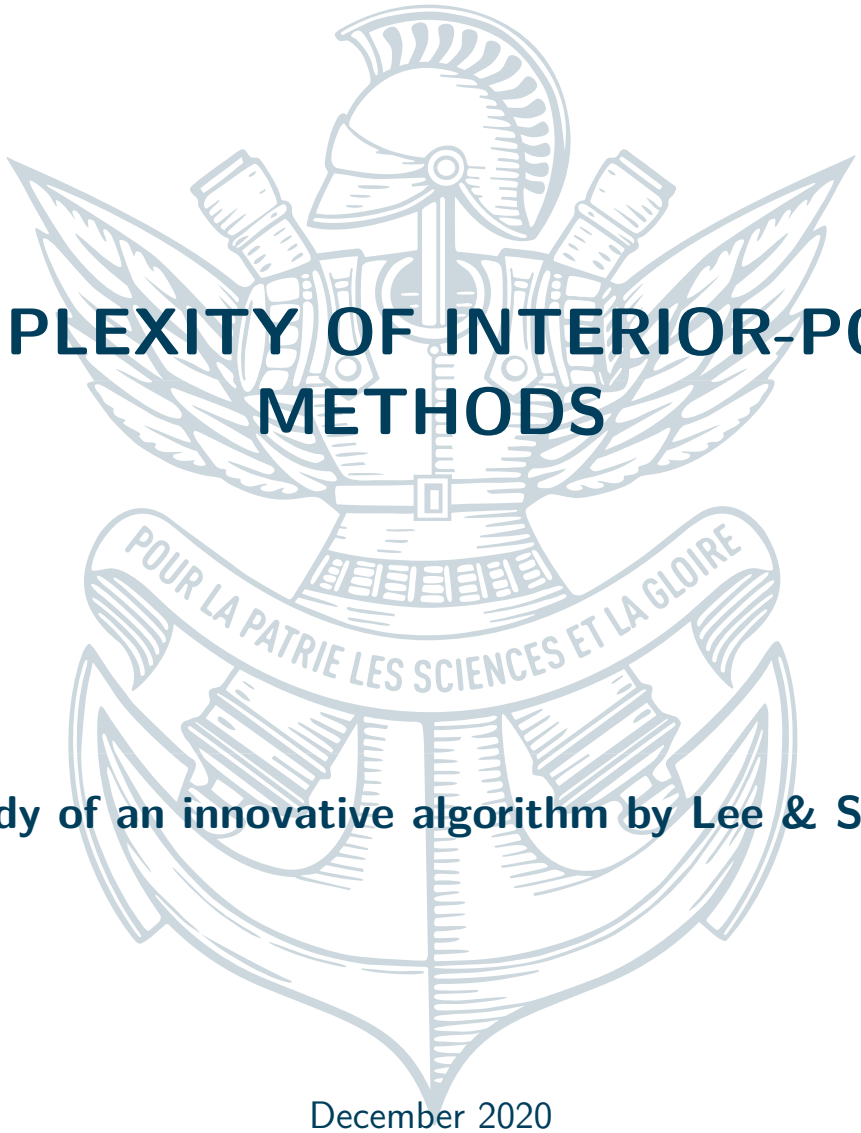


COMPLEXITY OF INTERIOR-POINT METHODS

A study of an innovative algorithm by Lee & Sidford

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ABSTRACT

Interior-point methods are a very important class of algorithms for linear programming. In this paper, we follow on a previous work by Xavier Allamigeon, Pascal Benchimol, Stéphane Gaubert and Michael Joswig [1] proving that log-barrier primal-dual interior-point methods are not strongly polynomial by presenting a counter-example. More precisely, we analyze an innovative algorithm proposed by Lee and Sidford [2], generalizing interior-point methods mainly by introducing "weighted path finding", with the objective to extend the results proven in [1] to this algorithm.

Our main finding is that a central path associated with a certain barrier-function that we determine lie in fact at the core of this algorithm. For instance, the last steps performed by the algorithm at the end for a fixed value of t converge towards a point of the central path (if it converges, which is the case in practice in this algorithm). Moreover, in an informal sense, throughout the execution of the algorithm, x remains close to this central path. We find a numerical scheme to approximate this central path, and our second important finding from numerical simulations is that the tropical central path is highly likely to be exactly the same as the one associated with the classical log-barrier. This opens the door to a proof of the non strongly polynomial complexity of this new algorithm by adapting the ideas of the proof of Allamigeon et al.

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1
INTRODUCTION AND CONTEXT

Let n , $m \in \mathbb{N}^*$ be two positive integers, $c \in \mathbb{R}^n$, $b \in \mathbb{R}^m$ two vectors and $A \in \mathbb{R}^{m \times n}$ a real $m \times n$ matrix. We are interested in solving linear programs which are optimization problems of the following form:

$$\text{minimize} \quad \langle c, x \rangle \quad \text{subject to } Ax \leq b \quad \text{and} \quad x \geq 0$$

where c is the cost vector and $x \geq 0$ mean that $x_i \geq 0$ for $1 \leq i \leq n$ (x is in the positive orthant).

By adding slack variables $w \in \mathbb{R}^m_{\geq 0}$ to x the linear program (P) can be transformed to the following form:

$$(P) \qquad \inf_{x \in \mathbb{R}^n, w \in \mathbb{R}^m} \langle c, x \rangle \quad \text{subject to } Ax + w = b \quad \text{and} \quad (x, w) \geq 0$$

We denote by $\mathcal{F}_P$ the feasible set of the problem (P) which is the set of vectors x, w that satisfy the constraints. Formally:

$$\mathcal{F}_p = \{(x, w) \in \mathbb{R}^n \times \mathbb{R}^m : Ax + w = b, x \geq 0, w \geq 0\}$$

We call the strictly feasible set $\mathcal{F}_P^s$ the subset of $\mathcal{F}_P$ that corresponds to strict positivity constraints over x and w:

$$\mathcal{F}_p^s = \{(x, w) \in \mathbb{R}^n \times \mathbb{R}^m : Ax + w = b, x > 0, w > 0\}$$

.

The dual program of (P) is denoted (D) and is written as follows:

$$(D) \qquad \sup_{y \in \mathbb{R}^m, s \in \mathbb{R}^n} \langle b, y \rangle \quad \text{subject to } A^T y + s = c \quad \text{and} \quad (y, s) \geq 0$$

This dual program also gives rise to the corresponding feasible and strictly feasible sets :

$$\mathcal{F}_D = \{(y, s) \in \mathbb{R}^m \times \mathbb{R}^n : A^T y + s = c,\ y \geq 0,\ s \geq 0\}$$

$$\mathcal{F}_D^s = \{(y, s) \in \mathbb{R}^m \times \mathbb{R}^n : A^T y + s = c,\ y > 0\ s > 0\}$$

We define the primal-dual feasible set $\mathcal{F}$ and strictly feasible set $\mathcal{F}^s$ as follows:

$$\mathcal{F} = \mathcal{F}_P \times \mathcal{F}_D \text{ and } \mathcal{F}^s = \mathcal{F}_P^s \times \mathcal{F}_D^s$$

A familiar result in linear programming (see [1] for more details) states that $(x, w) \in \mathbb{R}^n \times \mathbb{R}^m$ is solution to (P) if and only if there exist a couple $(y, s) \in \mathbb{R}^m \times \mathbb{R}^n$ such that :

$$\begin{cases} A^T y + s = c,\ s \geq 0,\ y \geq 0 \\ Ax + w = b,\ x \geq 0\ w \geq 0 \\ \langle x, s \rangle = 0 \end{cases} \tag{1}$$

The same result holds for the dual: (y, s) is an optimal solution of (D) if and only if there exist $(x, w) \in \mathbb{R}^n \times R^m$ such that (1) holds. In this case we denote by $z = (x, w, y, s)$ the primal-dual solution of (1).

2
LOG-BARRIER INTERIOR-POINT METHODS ARE NOT STRONGLY POLYNOMIAL

Interior-point methods are a very important class of algorithms for linear programming. Here, we briefly introduce them, give general background as well as key concepts to prove that log-barrier primal-dual interior-point methods are not strongly polynomial.

2.1 Barrier functions and interior-point methods

Positivity constraints are the main difficulty of the linear program summarized in (1). To get rid of these conditions, we penalize the points on the borders of the positive orthant by adding a barrier to the cost function that pushes the optimal solutions away from the boundaries of the feasible set $\mathcal{F}$ into the set $\mathcal{F}_s$. Formally, we define a family of programs parametrized by $\mu > 0$ as follows:

$$(P_\mu) \begin{cases} \inf \langle c, x\rangle + \mu.lb(x) \\ Ax + w = b, x \geq 0, w \geq 0 \end{cases}$$ where in many classical cases we define the barrier function with logarithms as such : $\mathrm{lb}(x) = \begin{cases} -\sum_{i=1}^{n} \log(x_i), \text{ if } x > 0 \\ +\infty, \text{ else} \end{cases}$

The dual of this problem is written as:

$$(D_\mu) \begin{cases} \sup \langle b, y\rangle - \mu \mathrm{lb}(s) \\ A^T y + s = c, y \geq 0, s \geq 0 \end{cases}$$

The *log-barrier* function lb, *pushes* the solution into the positive orthan by *penalizing* the objective function when we get close to the borders, as $\mathrm{lb}(x) \to +\infty$ when $x_i \to 0$. We can even "remove" the positivity constraints as they are implied by the form of the new objective function. The new task is simply to minimize a strictly convex function.

We could choose other barrier functions. The term barrier function has a precise definition given for example in [3]. Under the conditions of the definition, the following crucial theorem on central path holds.

2.2 Central path

Solving $(P_\mu) - (D_\mu)$ for all $\mu > 0$ gives an application:

$$C : \mu \in]0, +\infty[\to z(\mu) := (x(\mu), w(\mu), s(\mu), y(\mu)) \in \mathcal{F}^s$$

Notice that (P_μ) and (D_μ) becomes (P) and (D) when $\mu \to 0$ and that $(C(\mu))_\mu$ is an algebraic curve. For every barrier function (as defined in [3]) we have as we may expect $z(\mu) \xrightarrow[\mu \to 0]{} z^* = (x^*, w^*, y^*, s^*)$ where z^* is optimal primal-dual solution of the original problem. Therefore, the algebraic curve $(C(\mu))_{\mu \geq 0}$ is a path converging to the optimal solution of the problem as μ tends to 0 which is called the ***central path***. Interior-point algorithms consist of "following" this path to the optimal solution.

2.3 Background on the complexity of interior-point methods

In computational optimization, an open question asks whether linear programs can be solved with a strongly polynomial algorithm or not. This is known as Smale's 9th problem [4]. Strong polynomiality requires that :

1. the algorithm performs in polynomial time in the bit model (meaning that the execution time is polynomial in the number of bits of the input)
2. the number of arithmetic operations to be bounded by a polynomial in the number of numerical entries of the input, uniformly in their bit length.

The best known complexity performed by path-following methods so far, were obtained by Renegar [3] in $O(\sqrt{m}L)$ or more recently in [2] in $O(\sqrt{rank}L)$, where m is the number of inequality constraints, *rank* the rank of the constraints matrix and L is the total bit size of all coefficients. Hence, they are **polynomial in the bit model** since every iteration can be done in strongly polynomial time.

Smale's 9th problem raises the equivalent question of bounding the number of iterations by a polynomial depending only on the number of variables and constraints, regardless of their respective sizes.

In their article, Xavier Allamigeon and his co-authors showed that the log-barrier interior-point methods are not strongly polynomial by presenting a counter example instance inspired by a Game-Theory problem. Interestingly, the properties of the central path (and of the tropical central path) lie at the core of their proof. It is considered by many that the total curvature can be seen as an informal complexity measure of the central path. Intuitively, a central path with a large total curvature should be harder to approximate by linear segments.

The counterexample is the following. Given an integer number $r > 1$ and an Archimedean parameter $t_{puiseux}$ (we call it this way in reference to the field of Puiseux series, a very simple non-archimedean field, see next subsection for more details) we consider the following Linear Program $\mathbf{LW}_r(t_{puiseux})$:

$$\begin{array}{rl} \text{Minimize} & x_1 \\ \text{Subject to} & x_1 \leq t_{puiseux}^2 \\ & x_2 \leq t_{puiseux} \\ & x_{2j+1} \leq t_{puiseux}\, x_{2j-1}\,,\ x_{2j+1} \leq t_{puiseux}\, x_{2j}, \qquad 1 \leq j < r \\ & x_{2j+2} \leq t_{puiseux}^{1-1/2^j}(x_{2j-1} + x_{2j}), \qquad 1 \leq j < r \\ & x_{2r-1} \geq 0\,,\ x_{2r} \geq 0 \end{array} \tag{2}$$

$\mathbf{LW}_r(t_{puiseux})$ has $2r$ variables and $3r+1$ constraints and can be transformed to fit the form of (P) by adding slack variables $(w_j)_{1\leq j\leq r}$. When transformed, $\mathbf{LW}_r(t_{puiseux})$ has $5r+1$ variables and $3r+1$ constraints.

2.4 Field of Puiseux series and Tropical Geometry tools

We denote by $\mathbb{K}$ the field of real and absolutely convergent Puiseux series, ie: the field of elements $f = \sum_{\alpha\in\mathbb{R}} a_\alpha t^\alpha$ such that:

$$\begin{cases} (i)\ a_\alpha \in \mathbb{R} \text{ for all } \alpha \in \mathbb{R}, \\ (ii)\ supp(f) = \{\alpha \in \mathbb{R} : a_\alpha \neq 0\} \text{ is either finite or has } -\infty \text{as the only accumulation point} \end{cases} \tag{3}$$

The definition of the field $\mathbb{K}$ guarantees that every series $f = \sum_{\alpha\in\mathbb{R}} a_\alpha t^\alpha$ has a leading term $a_{\alpha_0} t^{\alpha_0}$ where alpha is the defined as $\alpha_0 := max(supp(f))$. This allows us to define a total ordering on $\mathbb{K}$ where an element is positive if its leading term has a positive coefficient α.

The map $val : \mathbb{K} \to \mathbb{R} \cup \{-\infty\}$ that allows us to establish a link between $\mathbb{K}$ (and ultimately the problem $\mathbf{LW}_r(t_{puiseux})$) and Tropical Geometry is defined by: $val(f)$ for $f \in \mathbb{K}$ is the greatest element α_0 of $supp(f)$ or equivalently:

$$val(f) = \lim_{t\to+\infty} \log_t |f(t)|$$

This is a non trivial *valuation* over the field of generalized Puiseux series that has the following basic properties:

$\forall f, g \in \mathbb{K}$:

$$val(f+g) \leq \max(val(f), val(g)) \quad \text{and} \quad val(fg) = val(f) + val(g)\,. \tag{4}$$

where the first inequality becomes an equality if the leading terms of f and g don't cancel which is the case when f, g are positive Puiseux series (i.e in $\mathbb{K}_+$). $val : (\mathbb{K}_+, +, \times) \to (\mathbb{T}, \oplus, \odot)$ where $\mathbb{T} := \mathbb{R} \cup \{-\infty\}$ is then a monotone and surjective semifield morphism from $\mathbb{K}_+$ to $\mathbb{T}$.

We can then define Puiseux polyhedron as we do in a real numbers setting (mainly thanks to Tarski's principle, see [1] for more details)

$$\mathcal{P} = \{x \in \mathbb{K}^d \colon Ax \leq b\}\,, \tag{5}$$

where $A \in \mathbb{K}^{p\times d}$, $b \in \mathbb{K}^p$ (with $p \geq 0$), and $\leq$ stands for the partial order over $\mathbb{K}^p$.

2.5 The tropical central path

As explained in [1], introducing the archimedean variable allows us consider the linear program $\mathbf{LW}_r(t_{puiseux})$ over the field $\mathbb{K}$ ($t_{puiseux}$ is seen as the formal Puiseux parameter), and then to compute the tropical central path $\mathcal{C}^{trop}(\lambda)$ which is defined as the image of the classical central path under the valuation map. In practice, we can compute the tropical central path by estimating the limit when $t_{puiseux} \to +\infty$ of the log in $t_{puiseux}$ basis of the point of the classical central path associated with $\mu = t^\lambda_{puiseux}$.

The combinatorial analysis of the tropical central path yields a lower bound on the curvature of the central path of the parametric family of linear problems over the reals $\mathbf{LW}_r(t_{puiseux})$, and also a lower bound on the number of iterations to perform to solve the linear program for log-barrier primal-dual interior-point methods.

The tropical central path with the classical log-barrier is given by the exact equations for the linear program $\mathbf{LW}_r(t_{puiseux})$ is for x :

$$
\begin{aligned}
x_1^\lambda &= \min(\lambda, 2) \\
x_2^\lambda &= 1 \\
x_{2j+1}^\lambda &= 1 + \min(x_{2j-1}^\lambda, x_{2j}^\lambda), \quad 1 \leq j < r \\
x_{2j+2}^\lambda &= (1 - 1/2^j) + \max(x_{2j-1}^\lambda, x_{2j}^\lambda), \quad 1 \leq j < r
\end{aligned}
$$

for the slack variables w :

$$
\begin{aligned}
& w_1^\lambda = 2,\ w_2^\lambda = 1,\ w_3^\lambda = 1 + \min(\lambda, 2) \\
\forall 1 < j < r \quad & w_{3j}^\lambda = 1 + \min(w_{3j-3}^\lambda, w_{3j-2}^\lambda) \\
\forall 1 \leq j < r \quad & w_{3j+1}^\lambda = 1 + w_{3j-1}^\lambda \\
& w_{3j+2}^\lambda = (1 - \frac{1}{2^j}) + \max(w_{3j} - 1, w_{3j-1}) \\
& w_{3r}^\lambda = \min(w_{3r-3}^\lambda, w_{3r-2}^\lambda) \\
& w_{3r+1}^\lambda = \max(w_{3r-3}^\lambda, w_{3r-2}^\lambda) - \frac{1}{2^{r-1}}
\end{aligned}
$$

This is a crucial point of the proof of Allamigeon et al. [1] that log-barrier interior-point methods are not strongly polynomial. Indeed, this tropical central path is made of at least 2^{r-1} tropical segments, making 2^{r-1} a lower bound of the number of iterations performed on this counterexample for sufficiently large values of $t_{puiseux}$.

2.6 Conclusion and goal of this paper

Lee & Sidford algorithm exposed in [2] is a generalization of interior-point methods. At first sight, it doesn't "follow" a clearly defined central path associated to a certain barrier function. The goal of this paper is to deepen the understanding this algorithm in order to hopefully extend the results of Allamigeon et al. to the Lee & Sidford algorithm.

3
AN INNOVATIVE ALGORITHM BY LEE & SIDFORD

3.1 Lewis weights

3.1.1 Motivation and definition

Lewis weights play a key role in the design and definition of both a self-concordant barrier function and the main algorithm of Lee & Sidford. In [2], the algorithm is designed to have an efficient complexity so the Lewis weights are never computed precisely. However, in this paper we will use a simpler, more computationally expensive, iterative algorithm for the Lewis weights presented in [5]. Lewis weights are fundamental in the theory of Banach spaces and a key tool for approximating a matrix in l_p norms. They generalize a fundamental l_2 measure of row importance known as leverage scores which are defined for $\mathbf{A} \in \mathbb{R}^{m\times n}$ as $\sigma(A) = \mathbf{diag}(A(A^TA)^{-1}A^T)$, *i.e.* the diagonals of the orthogonal projection matrix onto the image of A.

Intuitively, the $w_{l_p}(A)$ Lewis weight of a row i, denotes the importance of the ith row under l_p norm and it is known that sampling $O(n^{max(p/2,1)})$ rows of A with probability proportional to l_p Lewis weights and reweighting yields a matrix B such that with high probability $||Ax||_p \approx ||Bx||_p$ multiplicatively for all x. Recently, Cohen and Peng [5] studied Lewis weights in the context of solving l_p regression, showed that Lewis weights computation can be written as a convex optimization problem for $p \geq 2$, and provided a nearly constant iteration algorithm for computing Lewis weights for $p \in]0, 4[$.

Here is a formal definition of Lewis weights.

Definition 1. *For all $p > 0$ and non-degenerate matrix $A \in \mathbb{R}^{m,n}$ (i.e. A has full row rank) we define the l_p Lewis weight $w_p(A)$ as the unique vector $w \in \mathbb{R}^m_+$ such that $w = \sigma(\mathbf{W}^{\frac{1}{2}-\frac{1}{p}}A)$ where $\mathbf{W} = \mathbf{Diag}(w)$ and $\forall X \in \mathbb{R}^{m,n}$ $\sigma(X) = \mathbf{diag}(X(X^TX)^{-1}X^T) = \mathbf{diag}(P_X)$.*

3.1.2 Lewis weights through an optimization problem

Definition 2. *For non degenerate matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $p \neq 2$ we define the volumetric potential by*

$$\nu_p^A(w) = \frac{1}{1-\frac{2}{p}} \log\det(\mathbf{A}^T W^{1-\frac{2}{p}}\mathbf{A})$$

In [2] it is shown that for $p \neq 2$, $w_p(\mathbf{A})$ is the unique minizer of the equivalent convex optimization problems :

$$\min \nu_p^A(w) + \sum_{i\in[n]} w_i \text{ s.t } w \in \mathbb{R}^m_{>0} \text{ and } \min \nu_p^A(w) \text{ s.t } w \in \mathbb{R}^m_{>0}\,, \sum_{i\in[n]} w_i = n$$

This equivalent definition is central in the numerical scheme used in [2]. We base our simulation on the fixed point definition above.

3.1.3 An iterative algorithm to compute Lewis weights

Algorithm 1 Iteratively computing approximate Lewis weights

We first define the algorithm **LewisIterate(A, p,w)**.
w = **LewisIterate(A, p,w)**:

1. 1. For $i = 1 \dots n$
   (a) $\hat{\sigma}_i \leftarrow \sigma_i(W^{\frac{1}{2}-\frac{1}{p}}A)$ be a numerical approximation of the statistical leverage score of row i in $W^{\frac{1}{2}-\frac{1}{p}}A$
   (b) $w_i \leftarrow (w_i^{\frac{2}{p}-1}\hat{\sigma}_i)^{p/2})^p$
2. Return w

Then, we use this algorithm to define an iterative algorithm that computes approximate Lewis weights.
w = **ApproxLewisWeights(A, p, T)**

1. Initialize $w_i = 1$
2. For $t = 1 \dots T$ (a)
   (a) $w \leftarrow$ **LewisIterate**(A, p, w).
3. Return w.

A proof of the convergence of the iterative algorithm in case $p < 4$ is directly linked to the contraction property of the **LewisIterate** function and the Banach fixed point theorem. A more complete proof and bounds of errors are to be found in [5]. In practice, given the almost constant time of the convergence (exponentially fast given that $p = 1 - 1/log(4m)$), we chose $T = 10$.

3.2 Description and analysis of the algorithm

In their article [2], Lee & Sidford propose a new innovative algorithm for linear programming. In this paper, we look closely at this algorithm, ultimately aiming at proving or disproving its strong polynomial complexity. First, let's expose the motivation behind this algorithm.

3.2.1 Motivation

At the beginning of their paper, Lee & Sidford discuss their motivation behind their algorithm. It can be quickly summarized as follows :

- Nesterov and Nemirovski showed that given any ν-self-concordant barrier for the domain, an $O(\sqrt{\nu}L)$-iteration method could be achieved. Further, they showed that for any convex set in $\mathbb{R}^n$, there exists an $O(n)$-self-concordant barrier, called the universal barrier function. Therefore, in theory any such n-dimensional convex optimization problem can be solved in $O(\sqrt{n}L)$ iterations. However, this result is traditionally considered to be primarily of theoretical interest as the universal barrier function is difficult to compute.

- Lee & Sidford first wanted to find a $\tilde{O}(n)$-self-concordant barrier (the notation $\tilde{O}$ means that polylogarithmic factors are not taken into account), in order to use it to create a $\tilde{O}(\sqrt{n}L)$ iterations interior point method. They succeeded by finding such $\tilde{O}(n)$-self-concordant barrier based on Lewis weights.
- However, because of Lewis weights this barrier is still too long to compute to lead to an efficient algorithm.
- They overcame this obstacle by developing the "weighted path finding" algorithm, which is **at first sight totally independent from this motivation part**.

Indeed, **"weighted path finding"** is based on the observation that the weighting of each coordinate in the classical log-barrier can affect convergence. The idea is therefore to use the "weighted barrier function" $\phi(x) = \sum g_i(x)(-log(x_i))$ where g is a well-chosen weight function. A good choice for g, used by Lee & Sidford in their algorithm, is $g(x) = w_p(A_x) + c_0$ with $p = 1 - 1/(log(4m))$, $c_0 = n/(2m)$ and $A_x = X^{-1}A$, thanks to the nice properties of Lewis weights.

But again, it is still too long to compute since it requires to compute Lewis weights. Therefore, **they maintained separate the weights w and the current point x**. The main pattern of the algorithm is : update x , update w, update t , repeat.

The objective function to minimize is

$$f_{t,w}(x) = t\langle c, x\rangle + \sum w_i(-ln(x_i))$$

Throughout the algorithm, the weights w should remain "close" to the value $g(x)$, and x "close" to the minimum $f_{t,w}(x)$ with fixed w and t.

After deeper analysis (see subsection 3.3), however, it appears that this "weighted path finding" approach is in fact closely related to the new barrier function Lee & Sidford found.

3.2.2 Main properties

The main idea is to follow the classical path-following method for barrier $\phi(x) = \sum g_i(x)(-log(x_i))$. But the weights are separate from x for computational reasons and should be updated so as to remain "close" to $g(x)$. The main loop of the algorithm is update x , update w, update t , repeat.

This algorithm is proven to work through **loop-invariant**.

Keys concepts :

- Throughout the process, x should be close to its optimal value (i.e. to the central path associated with w). This is done through the study of **centrality** $\delta_t(x, w)$, formally defined page 12, which is a measure of the distance from x to the minimum of $f_t(x, w)$ for fixed w and t.
- $\Phi_\mu(x) = \sum(e^{\mu x_i} + e^{-\mu x_i})$ is used to measure the distance from w to $g(x)$, whose quantity should also remain small throughout the algorithm.

The main result is Theorem 40 in [2]. To make it short, it states that $\delta_t(x, w) \leq R$ AND that $\Phi_\mu(log(g(x)) - log(w)) \leq U$ is an invariant maintained provided that R and U are small enough.

3.2.3 Formal description of the algorithm

In order to initialize a good starting couple (x, w) (i.e. satisfying $\delta_t(x, w) \leq R$ and $\Phi_\mu(log(g(x)) - log(w)) \leq U$), Lee & Sidford propose the following trick (implemented in step 1 in the algorithm) :

- start with a point x in the domain
- calculate the weights accordingly, precisely enough so that $\Phi_\mu(log(g(x)) - log(w)) \leq U$ is satisfied
- carefully choose **another vector** d such that x is **exactly** on the weighted central path (choose $d_i = w_i/x_i$)
- the invariant conditions are therefore satisfied by (x, w) (with d instead of c)
- we use the procedure to **decrease** t
- when we change d in c, the invariant condition still holds provided that t is small enough
- so we have a couple (x, w) satisfying the invariant condition for vector c and we have initialized the process !

Here is a more formal description of the algorithm.

Algorithm 2 Outline of the algorithm of Lee & Sidford

We present here a synthetic vision of the algorithm proposed by Lee & Sidford.

1. **Toward the analytical center**. We begin with a vector x in the domain, and we compute corresponding weights w very close to $g(x)$. Then we carefully define a vector d such that x is **exactly** on the weighted central path. We start at $t = 1$ and until t is small enough we repeat :
   (a) One Newton step on x, i.e. we perform a Newton step aiming at minimizing the objective function with vector d and the barrier function (the weights are considered fixed) $f_{t,w}(.)$ conditioned by the fact that x belongs to the domain.
   (b) Computation of a new estimation of w.
   (c) $t = t(1 - \alpha)$.
2. **Increasing t**. We will to exactly the opposite process as in the first part but here **crucially** we use the **initial** vector c in the objective function. Until t is big enough we repeat :
   (a) One Newton step on x.
   (b) Computation of a new estimation of w.
   (c) $t = t(1 + \alpha)$.
3. **Last steps** t is now fixed, and we proceed to a few steps of alternatively updating x and w.

Output : x

3.3 Link with interior points methods

3.3.1 Convergence conditions for the last part of the algorithm

Let us start by characterising the convergence couple (x, w) of the step 3 of this algorithm. If the algorithm converges at $(x^\star, w^\star)$, then the newton step at that point is equal to 0 and $w_{x^\star} = g(x^\star) = w^\star$. The newton step h_{NEWTON} is defined by :

$$\arg\min_{A^T h=0} f_{t,w}(x) + \nabla f_{t,w}(x)^T h + \frac{1}{2} < h, H_x h >= h_{\text{NEWTON}}$$

Where H_x the hessian matrix of $f_{t,w}$ in x.

Since h_{NEWTON} is the solution of this optimization problem, with Lagrange multipliers for example we have $\nabla f_{t,w}(x) + H_x h_{\text{NEWTON}} = A\lambda$ with $\lambda \in \mathbb{R}^m$.

In the convergence of the algorithm, $h_{\text{NEWTON}} = 0$, and $\nabla f_{t,w}(x) = A\lambda$. Moreover, since we have equality constraints, x represents both the primal variable and its slack and $\nabla f_{t,w}(x) \in \mathbb{R}^{n+m}$ Recall the expression of $f_{t,w}(x) = c^T x - t\sum_{i=1}^{n} w_i \log(x_i) - t\sum_{i=n+1}^{n+m} w_i \log(x_i)$, where the last m variables are slack variables. Therefore, we have

$$(\nabla f_{t,w}(x))_i = c_i - t\frac{w_i}{x_i} \; \forall i \in [1:n]$$

and

$$(\nabla f_{t,w}(x))_i = -t\frac{w_i}{x_i} \; \forall i \in [n+1:n+m]$$

with

$$A^T = \left[\; A'^T \mid I_m \; \right]$$

Where A' is the original constraints matrix, before adding the slacks.

In the convergence of our numerical scheme we have $\nabla f_{t,w}(x^*) = A\lambda$ is consequently written

$$\lambda_i = -t\frac{w_i}{x_i^*} \; \forall i \in [n+1, n+m]$$
$$A'\lambda = c - t(\frac{w_i}{x_i^*})_{i\in[1,n]}.$$

3.3.2 Central path associated to another barrier function

Let us consider the minimization problem

$$\inf c^T x + t\Psi(x) \text{ s.t } x > 0, A^T x = b$$

with

$$\Psi(x) = \ln\det(A_x^T W_x^{1-\frac{2}{p}} A_x)$$

and $A_x = \mathbf{X}^{-1}A$, $\mathbf{X} = \text{Diag}(x)$, $W_x = \text{Diag}(w_p(x))$ the Lewis Weights matrix (cf part 2.1).

Let's define

$$f(x, w) = \ln\det(A_x^T W^{1-\frac{2}{p}} A_x)$$

We have then $\Psi(X) = f(x, w_x)$ and by the chain rule

$$\nabla_x \Psi = \nabla_x f(x, w_x) + \mathbf{J_w}(x)^T \nabla_w f(x, w_x)$$

The definition of the Lewis weights, as minimizer of a convex optimization problem, gives $\nabla_w f(x, w_x) = 0$

$$\nabla_x f(x, w_x) = \mathrm{Tr}((A_x^T W^{1-\frac{2}{p}} A_x)^{-1} \partial_x (A_x^T W^{1-\frac{2}{p}} A_x)) = \frac{w_x}{x}$$

A point $x^{(t)}$of the central path of the function Ψ is such that

$$\arg\min c^T x + t\Psi(x) \text{ s.t. } x > 0 \text{ and } A^T x = b = x^{(t)}$$

Thus $c^T x^{(t)} + t\nabla_x \Psi(x^{(t)}) = A\lambda$

with the same notation as in the previous subsection, considering A^T the constraint matrix as a $\mathbb{R}^{(n+m)\times m}$-matrix coming from the addition of slacks we find the same conditions on $x^{(t)}$:

$$\lambda_i = -t\frac{w_i}{x_i^{(t)}} \; \forall i \in [n+1, n+m]$$
$$A'\lambda = c - t(\frac{w_i}{x_i^{(t)}})_{i\in[1,n]}.$$

Thus, under the assumption that the last step of the algorithm converges say towards $(x^\star, w^\star)$, we have shown that $x^\star$ belongs to the central path associated with Ψ.

3.4 New objectives

Throughout the algorithm, x stays informally "near" the central path associated with Ψ (in the sense that x is close to the central path associated with w, w is close to $g(x)$, and that if both distances were zero x would be on the central path).

So, informally, the algorithm appears to be a kind of path-following algorithm. Moreover, we can create an algorithm that will numerically compute this central path, based on the observation of the previous subsection.

Our objective is now to implement such an algorithm, in order to get an insight of how this central path looks. In particular, if the tropical central path on the counterexample proposed by Allamigeon et al. in [1] is the same as the tropical central path associated with the classical central path, it would give hope in the possibility of extending this result and prove that Lee & Sidford algorithm is not strongly polynomial.

4
NUMERICAL SIMULATIONS

4.1 Description of a numerical scheme for the central path

We present here a numerical scheme, that, when it converges, converges to the central path.

Basically, we want to perform as many iterations as we want for a fixed value of t, like in the last step of Lee & Sidford algorithm, hopping it will converge. We can define the sopping condition as we want (see ϵ below). However, we observed that **it doesn't always converge**. For a fixed t, the iterated convergence scheme can sometimes be attracted to two different values (x_1, w_1) and (x_2, w_2) that send to each other. Worse, the Newton steps can break the positivity constraint. For those reasons, we would like to be "close enough" to the actual central path to avoid those problems, and use the theoretical guarantees proved by Lee & Sidford to have a rigorous numerical scheme. That is the reason why we adapted Lee & Sidford algorithm, based on the invariant loop that this quantity remains "small" throughout the algorithm.

Nevertheless, the value of α they used was too small for computation in a reasonable time. So we performed our idea with a much bigger value of α, typically 0.1, and introduced simulation parameters λ_1, λ_2 and u_{bound} to solve the convergence problem.

Before describing the algorithm, let's introduce its inputs :

- $Time$: a list containing all the values of parameter t for which we want to get an estimation of the corresponding points on the central path.
- t_1 : in the least value of t we will reach in the second step of our numerical sheme (when t decreases)
- α : the multiplicative steps will be $1 - \alpha$ in the decreasing phase (step 2) and $1 + \alpha$ in the increasing one (step 3)
- λ_1 : instead of doing $x = x + h$ where h is the Newton step, we do $x = x + \lambda_1 h$ in step 2 and 3
- λ_2 : $x = x + \lambda_2 h$ in step 4 (we used a different value because at this point of the algorithm we face different problems).
- u_{bound} : if, during the decreasing phase, the L_1-norm of h is strictly greater than u_{bound}, we do iterations t being fixed, like in step 4. We don't use it in practice, setting a very high value for u_{bound}. Even though this condition is not activated, it can be seen as guarantee from exploding steps.
- ϵ : Stopping condition for step 4, and directly linked to the precision of the output.

Algorithm 3 Numerical scheme to approximate the central path

We present here a synthetic vision of the algorithm we designed to approximately compute the central path.

Input : $Time$, t_1, α, λ_1, λ_2, u_{bound}, ϵ.

1. **Computation of an initial point**. We implemented a procedure to output a vector x belonging to the domain.

2. **Toward the analytical center**. We begin with this vector x in the domain, and we compute corresponding weights w (via the Lewis weights computation procedure, see $g(x)$ and Lewis weights computation). Then we define the vector d such that x is **exactly** on the weighted central path, exactly like Lee & Sidford algorithm. We start at $t = 1$ and while $t > t_1$ we repeat :

   (a) One Newton step on x **scaled** by λ_1. (d in the objective function)

   (b) Computation of a new estimation of w.

   (c) $t = t(1 - \alpha)$.

3. **Increasing t**. We will do exactly the opposite process as in the first part but here **crucially** we use the **initial** vector c in the objective function. Until t is greater than a new value in $Time$, we do :

   (a) One Newton step on x scaled by λ_1. (c in the objective function)

   (b) Computation of a new estimation of w.

   (c) $t = t(1 + \alpha)$.

   And we go to the next step.

4. **Last steps for a given value of t** t is now fixed at the value in Time we are considering, and we proceed to steps of alternatively updating x (Newton step on x **scaled** by λ_2) and w until the L_1-norm of the Newton step on x is smaller than ϵ. We store the value of x. If all values in the list $Time$ have not been considered, we go back to the previous step.

Output : Approximations of the points of the central path corresponding to the values of t in the list $Time$.

4.2 Numerical outputs

4.2.1 On a simple linear program

First, we applied our numerical scheme to a simple linear program to make tests and comparisons with the central path associated to the log-barrier. This linear program is

$$\begin{aligned} \text{Minimize} \quad & x_1 + t_{puiseux}^3 x_2 \\ \text{Subject to} \quad & x_1 + x_2 \leq 2 \\ & t_{puiseux} x_1 \leq 1 + t_{puiseux}^2 x_2 \\ & t_{puiseux} x_2 \leq 1 + t_{puiseux}^3 x_1 \\ & x_1 \leq t_{puiseux}^2 x_2 \\ & x_1, x_2 \geq 0 \end{aligned} \tag{6}$$

Here is what we obtained :

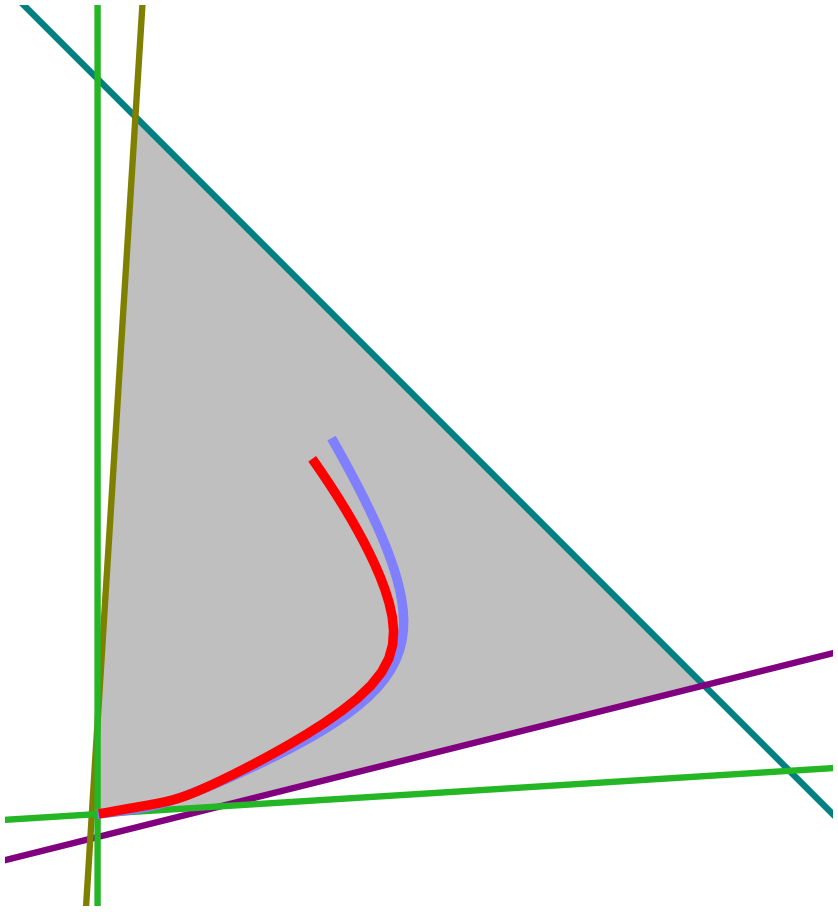


Figure 1: Central path with respect to Lee & Sidford barrier computed with our numerical scheme (in red) and with respect to the classical log-barrier (in blue) for $t_{puiseux} = 4$. The grey area corresponds to the interior of the domain of 6.

Numerical parameters : $t_1 = 10^{-30}$, $\alpha = 0.1$, $\lambda_1 = 10^{-4}$, $\lambda_2 = 10^{-1}$, $u_{bound} = 10^8$, $\epsilon = 10^{-3}$.

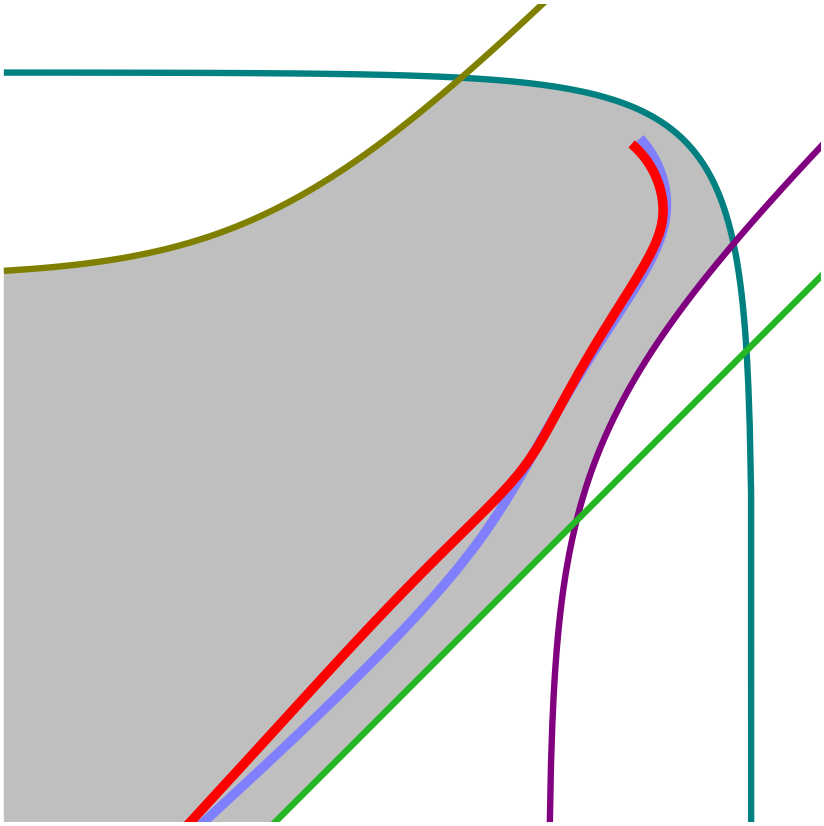


Figure 2: Log in $t_{puiseux}$ basis of the central path with respect to Lee & Sidford barrier computed with our numerical scheme (in red) and with respect to the classical log-barrier (in blue) for $t_{puiseux} = 4$. The grey area corresponds to the interior of the domain of 6.
Numerical parameters : $t_1 = 10^{-30}$, $\alpha = 0.1$, $\lambda_1 = 10^{-4}$, $\lambda_2 = 10^{-1}$, $u_{bound} = 10^8$, $\epsilon = 10^{-3}$.

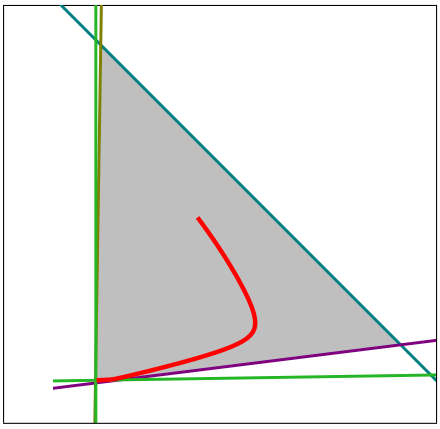


Figure 3: $t_{puiseux} = 8$
Numerical parameters : $t_1 = 10^{-30}$, $\alpha = 0.1$, $\lambda_1 = 10^{-4}$, $\lambda_2 = 10^{-1}$, $u_{bound} = 10^8$, $\epsilon = 10^{-5}$.

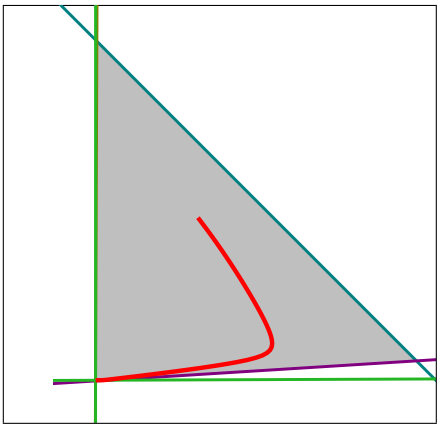


Figure 4: $t_{puiseux} = 16$
Numerical parameters : $t_1 = 10^{-30}$, $\alpha = 0.1$, $\lambda_1 = 10^{-4}$, $\lambda_2 = 10^{-1}$, $u_{bound} = 10^8$, $\epsilon = 10^{-5}$.

To put it in a nutshell, we observe on this very simple linear program that the central path with respect to the new Lee & Sidford barrier is very close to central path with respect to the classical

log-barrier, though they are clearly distinct in particular near the analytical center (the point of the central path corresponding to $\mu = +\infty$). We also note from 3 and 4 that the curvature of the classical central path increases as we increase $t_{puiseux}$.

Then, we apply our numerical scheme to the counterexample of Xavier Allamigeon et al. to get numerical approximations of the tropical central path.

4.2.2 On the counterexample

Recall that given an integer number $r > 1$ and an Archimedian parameter $t_{puiseux}$ the following Linear Program $\mathrm{LW}_r(t_{puiseux})$ has been used by Xavier Allamigeon and his co-authors to disprove strong polynomial complexity. One key argument of their proof was to show that for this particular linear program the tropical central path (with respect to the log-barrier) has a precise shape.

$$
\begin{array}{rll}
\text{Minimize} & x_1 & \\
\text{Subject to} & x_1 \leq t_{puiseux}^2 & \\
& x_2 \leq t_{puiseux} & \\
& x_{2j+1} \leq t_{puiseux}\, x_{2j-1}\,,\ x_{2j+1} \leq t_{puiseux}\, x_{2j}, \qquad 1 \leq j < r & \mathrm{LW}_r(t_{puiseux}) \\
& x_{2j+2} \leq t_{puiseux}^{1-1/2^j}(x_{2j-1} + x_{2j}), \qquad 1 \leq j < r & \\
& x_{2r-1} \geq 0\,,\ x_{2r} \geq 0 &
\end{array}
$$

$\mathbf{LW}_r(t_{puiseux})$ has $2r$ variables and $3r + 1$ constraints and can be transformed to fit the form of (P) by adding slack variables $(w_j)_{1 \leq j \leq r}$. When transformed, $\mathbf{LW}_r(t_{puiseux})$ has $5r + 1$ variables and $3r + 1$ constraints.

$\mathbf{LW}_r(t_{puiseux})$ can be rewritten in a matricial form as follows:

$$
\begin{array}{rl}
\text{Minimize} & c.x \\
\text{Subject to} & A(t_{puiseux})x \leq b(t_{puiseux}) \\
& x \geq 0
\end{array}
$$

Very importantly, our numerical simulations, with the proof of subsection 3.3, very strongly suggest that the tropical central path (with respect to the Lee & Sidford barrier) has exactly the same shape for this linear program. It gives hope in the idea to generalize Xavier Allamigeon et al. proof to Lee & Sidford algorithm by using the same counterexample and studying its tropical central path.

Indeed, here is what we obtained.

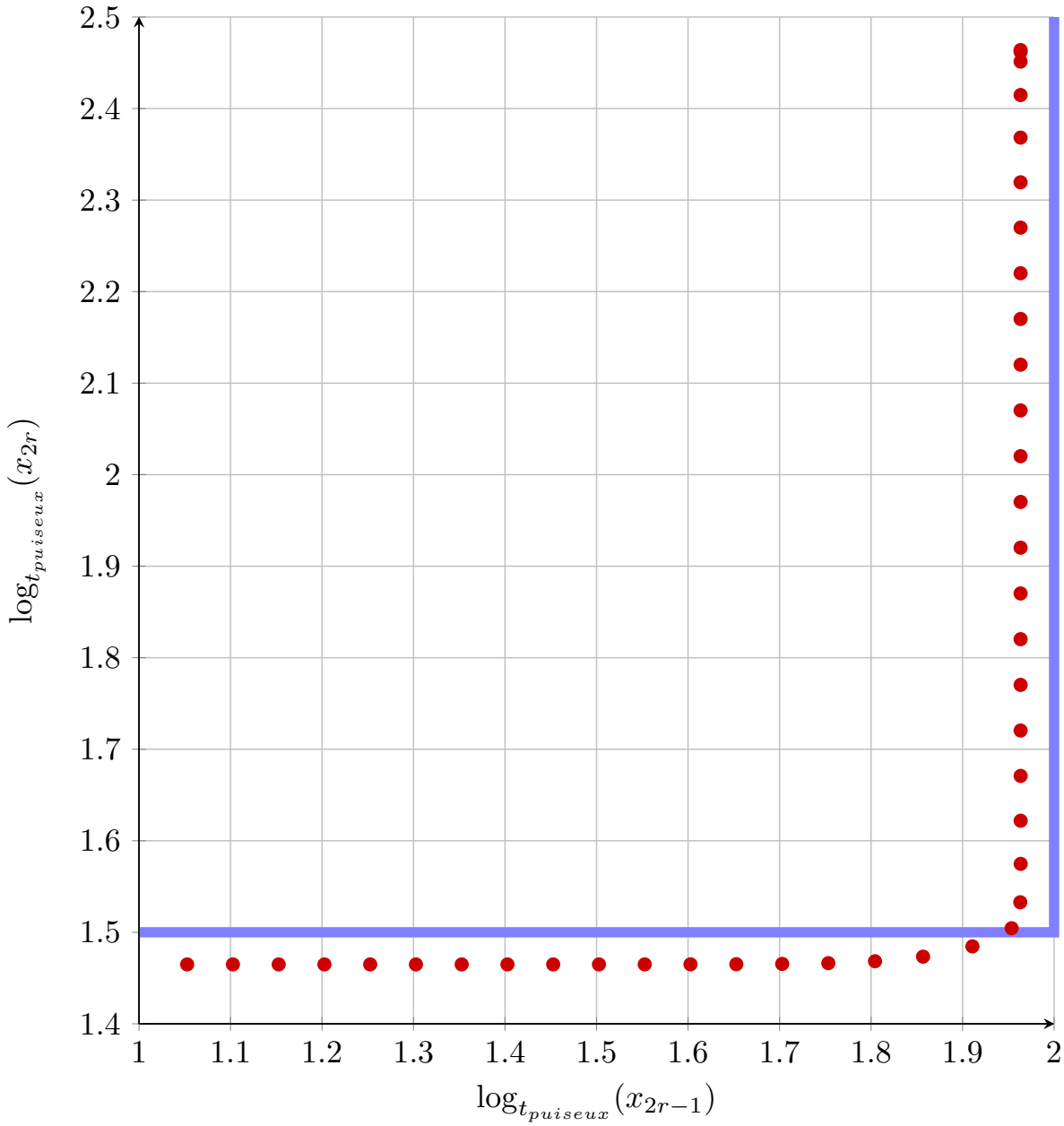


Figure 5: In red : log-central path for $r = 2$, $t_{puiseux} = 10^8$. In blue : the tropical central path associated with the classical log-barrier.

Numerical parameters : $t_1 = 10^{-40}$, $\alpha = 0.05$, $\lambda_1 = 10^{-7}$, $\lambda_2 = 10^{-1}$, $u_{bound} = 10^{10}$, $\epsilon = 10^{-7}$.

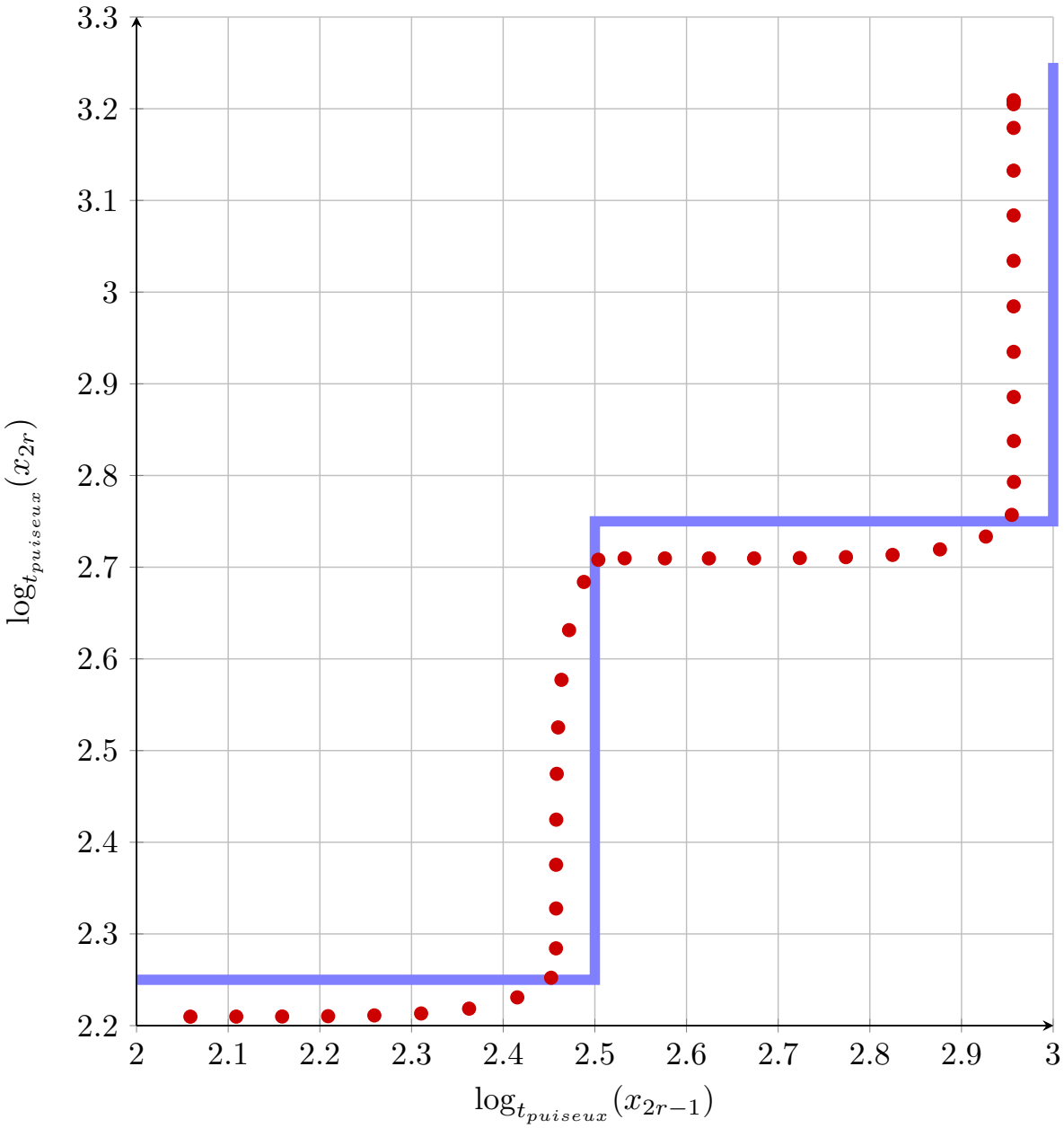


Figure 6: In red : log-central path for $r = 3$, $t_{puiseux} = 10^8$. In blue : the tropical central path associated with the classical log-barrier.

Numerical parameters : $t_1 = 10^{-40}$, $\alpha = 0.05$, $\lambda_1 = 10^{-7}$, $\lambda_2 = 10^{-1}$, $u_{bound} = 10^{10}$, $\epsilon = 10^{-7}$.

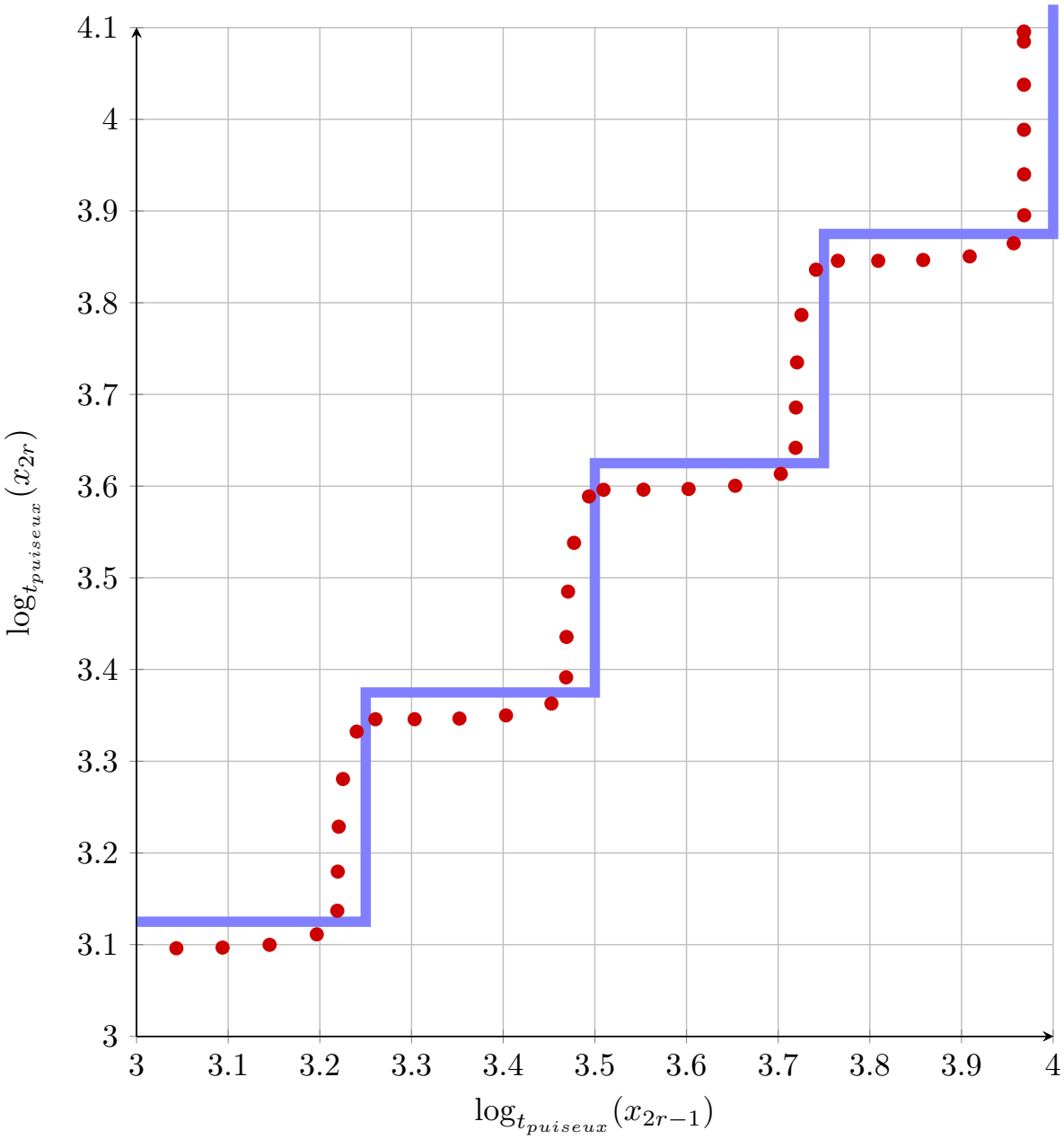


Figure 7: In red : log-central path for $r = 4$, $t_{puiseux} = 10^{12}$. In blue : the tropical central path associated with the classical log-barrier.

Numerical parameters : $t_1 = 10^{-40}$, $\alpha = 0.05$, $\lambda_1 = 10^{-7}$, $\lambda_2 = 10^{-1}$, $u_{bound} = 10^{15}$, $\epsilon = 10^{-7}$.

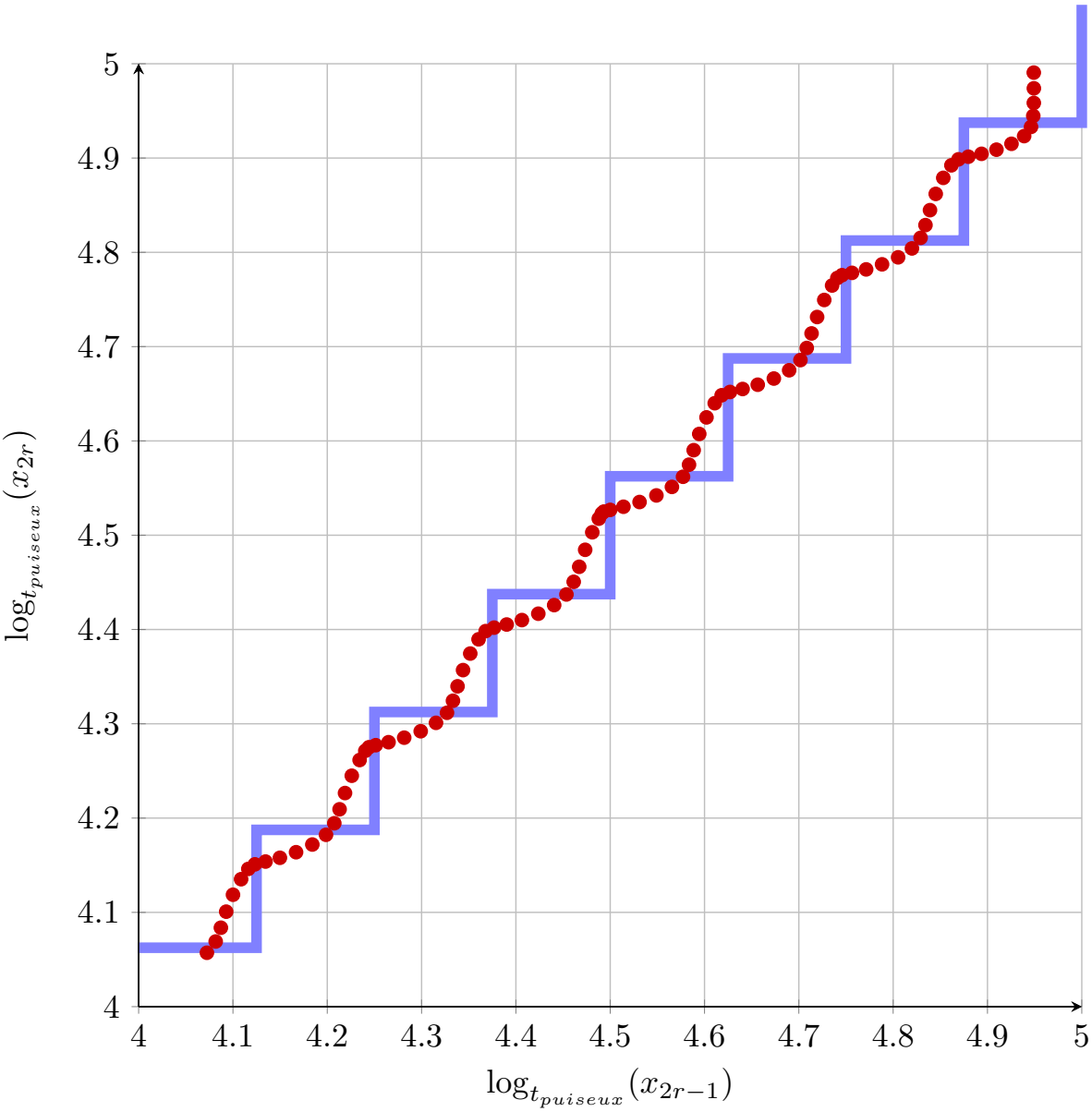


Figure 8: In red : log-central path for $r = 5$, $t_{puiseux} = 10^8$. In blue : the tropical central path associated with the classical log-barrier.

Numerical parameters : $t_1 = 10^{-40}$, $\alpha = 0.05$, $\lambda_1 = 10^{-7}$, $\lambda_2 = 10^{-2}$, $u_{bound} = 10^{20}$, $\epsilon = 10^{-7}$.

Note : for $r = 5$, convergence toward the tropical central path is probably achieved for greater values of $t_{puiseux}$ but we couldn't obtain it because it was too time-consuming.

5
CONCLUSION AND WHAT REMAINS TO BE DONE

In this project, we gave a mathematical proof that a numerical scheme, when it converges, converges towards a central path with respect to a new identified barrier function. We implemented this numerical scheme, and the numerical simulations we obtained very strongly suggest that the tropical central path is the same as for the classical log-barrier. Moreover, this central path associated with this barrier-function lie in fact at the core of Lee & Sidford algorithm. For instance, the last steps performed by the algorithm at the end for a fixed value of t converge towards a point of the central path (if it converges, which is the case in practice). Furthermore, in an informal sense, throughout the execution of the algorithm, x remains close to this central path. Therefore, we can hope to extend the result proven by Xavier Allamigeon et al. in [1] and prove that the algorithm proposed by Lee & Sidford is not strongly polynomial.

In order to do so, we would have to :

- Prove that the tropical central path is indeed the same as the tropical central path associated with the log-barrier.
- Carefully control the distance between x and the central path throughout the algorithm so that the lower bound on the number of steps applies to the new numerical scheme as well.

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